Riphah HireHive

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2. Software Requirements

1. User Registration and Login

- SRS-1.1: The system shall allow FYP students, volunteers, and 2nd to 6th semester students to register using a valid Riphah email address (for example 54910@students.riphah.edu.pk).
- SRS-1.2: The system shall reject a registration when the email address is not in the Riphah format.
- SRS-1.3: The system shall verify that the email address and SAP ID are unique before creating an account.
- SRS-1.4: The system shall send a verification code to the registered email address.
- SRS-1.5: The system shall not allow a user to log in before completing account verification.
- SRS-1.6: The system shall encrypt user passwords before storing them in the database.

2. User Management and Roles

- SRS-2.1: The system shall allow the admin team to create, edit, deactivate, and delete user accounts.
- SRS-2.2: The system shall show each user only the screens and data that are allowed for their role.
- SRS-2.3: The system shall allow the event organizer to give selected team members the right to edit shared forms such as the demand form.
- SRS-2.4: The system shall generate a username and password for each selected volunteer and send it to them.

3. Profile and CV Management

- SRS-3.1: The system shall allow users to update their name, phone number, and profile picture.
- SRS-3.2: The system shall allow users to change their password after verifying their existing password.
- SRS-3.3: The system shall allow FYP students and 2nd to 6th semester students to upload and replace their CV.
- SRS-3.4: The system shall display an FYP student's name, email, SAP ID, project details, and supervisor on their portal.
- SRS-3.5: The system shall tell a student when required profile information or the CV is missing.

4. Event Management

- SRS-4.1: The system shall allow the admin team to create an open house event with its date, time, venue, and registration deadlines.
- SRS-4.2: The system shall provide registration links for FYP teams, volunteers, students, and companies.
- SRS-4.3: The system shall provide a demand form for tables, chairs, sound system, microphones, and decoration theme.
- SRS-4.4: The system shall export the approved demand form as a PDF.
- SRS-4.5: The system shall collect each team member's free and busy time and suggest a common free slot for meetings.

5. FYP Team and Project Registration

- SRS-5.1: The system shall allow FYP students to submit team details, including the number of members, member names, and SAP IDs.
- SRS-5.2: The system shall allow FYP students to submit the supervisor name, technology used, and project description.
- SRS-5.3: The system shall prevent the same SAP ID from being registered in more than one team.
- SRS-5.4: The system shall warn the team when required information, such as the supervisor name, is missing.
- SRS-5.5: The system shall allow supervisors to approve FYP team registrations.
- SRS-5.6: The system shall allow the admin team to view and export the list of registered teams by program (BCA, SE, CS).

6. Volunteer Registration and Selection

- SRS-6.1: The system shall allow students to register as volunteers by entering their name, SAP ID, WhatsApp number, and experience.
- SRS-6.2: The system shall not accept volunteer registrations after the deadline.
- SRS-6.3: The system shall shortlist volunteers based on experience and participation in previous open houses.
- SRS-6.4: The system shall suggest suitable volunteers for each duty, such as social media or welcoming.
- SRS-6.5: The system shall display each selected volunteer's duties, placement, responsibilities, and rules on their portal.

7. Company Registration and Invitation

- SRS-7.1: The system shall allow the admin team to add and store company records, including company name, contact person, email, and phone number.
- SRS-7.2: The system shall send an invitation email with the invitation card, student information, table details, and attachments.
- SRS-7.3: The system shall allow companies to confirm attendance with the number of people and the name of the team leader.

8. Table Allocation and Tags

- SRS-8.1: The system shall assign a unique table number to every project.
- SRS-8.2: The system shall assign table numbers in one continuous sequence across programs (for example, if BCA uses tables 1 to 8, SE starts at table 9).
- SRS-8.3: The system shall prevent two projects from having the same table number.
- SRS-8.4: The system shall generate a printable table tag containing the project name, SAP ID, email, supervisor, technology, program, and table number.
- SRS-8.5: The system shall allow the admin team to review and correct a tag before printing.
- SRS-8.6: The system shall display the table number and location to each FYP team on their portal.

9. Seminar and Attendance

- SRS-9.1: The system shall allow the admin team to schedule a seminar with a room number and time.
- SRS-9.2: The system shall notify students, volunteers, teachers, and the admin team of the seminar room and time on their portals.
- SRS-9.3: The system shall generate an attendance sheet for registered FYP students and volunteers with SAP ID, name, and role.
- SRS-9.4: The system shall allow the admin team to mark seminar attendance.
- SRS-9.5: The system shall notify students who did not attend the seminar that they are not eligible for the open house.
- SRS-9.6: The system shall allow the admin team to export the attendance sheet.

10. LCD Issue and Return

- SRS-10.1: The system shall allow FYP groups to request an LCD from the lab.
- SRS-10.2: The system shall fill the group name, member names, SAP IDs, and supervisor name in the LCD request from the project record.
- SRS-10.3: The system shall display the disclaimer that the group bears any damage, and shall require the group to accept it before the request is submitted.
- SRS-10.4: The system shall show lab staff a list of all LCD requests.
- SRS-10.5: The system shall allow lab staff to record the issue and return of the LCD, HDMI cable, and power cable, with a note for any damage.
- SRS-10.6: The system shall keep a record of LCD issue and return for every open house and list the groups that have not returned the LCD.

11. QR Code and Check-in

- SRS-11.1: The system shall generate a unique QR code for each invited company.
- SRS-11.2: The system shall allow volunteers at the welcome desk to check in a company by scanning its QR code or ticking its checkbox.
- SRS-11.3: The system shall allow volunteers to search for a company by company name or person name.
- SRS-11.4: The system shall show volunteers the list of expected companies with company name, number of people, and main person.
- SRS-11.5: The system shall allow volunteers to add a new entry for a company that is not registered and mark it present.
- SRS-11.6: The system shall save every check-in record and allow it to be exported.

12. Job Fair and Interview Management

- SRS-12.1: The system shall allow companies to search CVs by keyword, skill, or program.
- SRS-12.2: The system shall show whether a student's skills and experience match what the company is looking for.
- SRS-12.3: The system shall allow companies to shortlist students.
- SRS-12.4: The system shall allow companies to schedule interviews with shortlisted students.
- SRS-12.5: The system shall notify students about their interview schedule.

13. FYP Exhibition and Evaluation

- SRS-13.1: The system shall allow companies and guests to browse FYP projects and filter them by technology or program.
- SRS-13.2: The system shall allow companies to save a shortlist of projects.
- SRS-13.3: The system shall allow FYP teams to book a presentation room and time slot.

14. Volunteer Task Management and Tracking

- SRS-14.1: The system shall allow the admin team to assign duties and tasks to volunteers with a priority.
- SRS-14.2: The system shall show each volunteer a to-do checklist and allow them to mark tasks as done.
- SRS-14.3: The system shall remind volunteers about tasks that are not yet completed.
- SRS-14.4: The system shall allow volunteers to check in and check out, and record the time.
- SRS-14.5: The system shall allow the admin team to view volunteers' placement and task progress, and to send reminder messages to them.
- SRS-14.6: The system shall allow volunteers to submit a short report of their work and allow the admin team to give appreciation for good performance.

15. Search and AI Matching

- SRS-15.1: The system shall allow companies to enter the kinds of projects they are interested in.
- SRS-15.2: The system shall recommend matching projects and their table numbers to a company based on its interests.
- SRS-15.3: The system shall allow users to search FYP projects by keyword, technology, or program.
- SRS-15.4: The system shall flag duplicate or invalid records, such as the same SAP ID twice, a wrong email domain, or a missing supervisor, for admin review.
- SRS-15.5: The system shall allow companies to mark a project as "interested" or "interview later".

16. Notifications and Announcements

- SRS-16.1: The system shall send notifications to users inside the application and by email.
- SRS-16.2: The system shall send reminders about deadlines, the seminar, and the event day.
- SRS-16.3: The system shall allow the admin team to send one notification to a whole group, such as all volunteers, all FYP students, or all companies.
- SRS-16.4: The system shall allow the admin team to send announcements on the event day, such as a room change.
- SRS-16.5: The system shall allow users to manage their notification preferences.
- SRS-16.6: The system shall keep a history of notifications sent.

17. Certificates

- SRS-17.1: The system shall generate an e-certificate or experience letter for volunteers after the event.
- SRS-17.2: The system shall base a volunteer's certificate on their recorded attendance and duty hours.

- SRS-17.3: The system shall allow the admin team to approve certificates before they are released.
- SRS-17.4: The system shall allow users to download their certificates from their portal.
- SRS-17.5: The system shall keep a record of all certificates issued.

18. Analytics and Reports

- SRS-18.1: The system shall display a dashboard of registrations, volunteers selected, tables assigned, companies confirmed, and attendance.
- SRS-18.2: The system shall alert the admin team about problems, such as companies not yet confirmed or tables with no project.
- SRS-18.3: The system shall allow the admin team to export lists and reports to Excel and PDF.
- SRS-18.4: The system shall generate a post-event report of attendance, companies visited, and volunteer performance.
- SRS-18.5: The system shall keep the data of previous open houses so that the next event can use it.
- SRS-18.6: The system shall show the number of company visits for each project.

3. Non-Functional Requirements

1. Usability

- UR-1.1: A first-time FYP student shall be able to complete team registration within 10 minutes without external help.
- UR-1.2: A first-time volunteer shall be able to complete the volunteer registration form within 5 minutes without external help.
- UR-1.3: A volunteer at the welcome desk shall be able to check in a company within 1 minute, by QR scan or by search.
- UR-1.4: A user shall be able to reach their main information (for example table number, duties, or recommended projects) within 2 interactions from the home screen.
- UR-1.5: The system shall prevent a form from being submitted when required fields are empty, show an error message that explains how to fix it (for example the correct Riphah email format), and show a confirmation after every successful submission.
- UR-1.6: The system shall use simple words, the same terms, and the same navigation on all screens, allow text size to be increased up to 200%, and not use color alone to show important information.
- UR-1.7: At least 80% of representative test users shall rate the system as satisfactory or higher in a post-test survey.

2. Performance

- PR-1.1: Each page shall load within 3 seconds under normal use.
- PR-1.2: Search results for CVs, projects, and companies shall appear within 3 seconds.
- PR-1.3: A check-in record shall be saved within 2 seconds.
- PR-1.4: The system shall support at least 300 users at the same time on the event day without errors.
- PR-1.5: The system shall generate table tags for all projects within 1 minute.
- PR-1.6: The system shall send a notification or email to a whole group within 5 minutes.

3. Security

- SEC-1.1: The system shall authenticate every user before giving access to any data.
- SEC-1.2: The system shall store passwords in encrypted form only.
- SEC-1.3: The system shall lock an account for a short time after 5 failed login attempts.
- SEC-1.4: The system shall not allow a user to open another role's screens or data, even by changing the web address.
- SEC-1.5: The system shall show a student's CV and phone number only to authorized users, and to companies only when the student has allowed it.
- SEC-1.6: The system shall transfer all data over an encrypted connection (HTTPS).
- SEC-1.7: The system shall log out a user automatically after 30 minutes of inactivity.
- SEC-1.8: The system shall accept only allowed file types (for example PDF and Word) and a maximum file size when users upload CVs and documents.
- SEC-1.9: The system shall use student and company data only for open house purposes.

4. Reliability and Availability

- REL-1.1: The system shall be available at least 99% of the time during the registration period and on the event day.
- REL-1.2: The system shall take an automatic backup of all data every day.
- REL-1.3: The system shall be able to restore data from the latest backup within 2 hours.
- REL-1.4: The system shall not lose already saved data if the internet or server fails.
- REL-1.5: The system shall show a clear error message and keep working when a service such as email is unavailable.
- REL-1.6: The system shall not create duplicate records when a user submits the same form twice.

5. Compatibility and Portability

- COM-1.1: The system shall work on the latest versions of Chrome, Edge, Firefox, and Safari.
- COM-1.2: The system shall adjust its screens to phone, tablet, and laptop sizes.
- COM-1.3: The exported Excel and PDF files shall open correctly in standard Excel and PDF readers.
- COM-1.4: The printable table tags and PDFs shall print correctly on A4 paper.
- COM-1.5: The generated QR codes shall be readable by a standard phone camera.

6. Scalability

- SCA-1.1: The system shall allow the number of FYP projects, volunteers, and companies to grow in later open houses without redesign.
- SCA-1.2: The system shall store the data of multiple open houses separately.
- SCA-1.3: The table numbering shall keep working correctly when more programs or projects are added.

7. Maintainability

- MNT-1.1: The admin team shall be able to change deadlines, email templates, and the evaluation rubric without changing the code.
- MNT-1.2: The system shall be built in separate modules so that one module can be updated without affecting the others.
- MNT-1.3: The system shall record errors in a log for troubleshooting.
- MNT-1.4: The project shall provide a user manual for the admin team and volunteers.

4. UseCase Diagram

